

Number Theoretic Functional Equations

Timothy Yau

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1 References

1. Functions, Intermediate Notes, Chapter 4
2. Functional Equation, Pang Cheng Wu, Chapter 5

2 Review of Basics

When treating functional equations, the most common type of problems involve finding functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Some variants include finding functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$, $f : \mathbb{Z} \rightarrow \mathbb{Z}$ and $f : \mathbb{N} \rightarrow \mathbb{N}$, with the condition still being an equality. Despite the change in the domain and codomain of the function, the techniques more or less remain the same as the ones described last time. The only slight change is that induction alone will be sufficient to pin down all function values. In fact, they are usually classified as A in the IMO shortlist. (See for example IMO shortlist 2014 A4 and IMO shortlist 2019 A7).

Meanwhile, there are certain classes of equations with a divisibility condition. They are more number theoretic in nature, and are usually classified as N in the IMO shortlist.

The approach to this class of problems is different from those introduced on 15 July. It comes to two ways to interpret (or make use of) the statement $a \mid b$.

1. Firstly, we can interpret $a \mid b$ in terms of primes. We have

$$a \mid b \iff v_p(a) \leq v_p(b)$$

for all prime p . (Recall that $v_p(a)$ is the exponent of p in the prime factorization of a .)

We can also adopt the weakening principle (a term invented by myself) For example, we have

- (a) If $p \mid a$ then $p \mid b$.
- (b) If b is a prime then $a = \pm b$ or $a = \pm 1$.

In particular, prime numbers play an important role. Therefore, it is often useful to pin down the prime values. This is best explained by Arvin Leung, in the book IMO shortlist compilation, P.145.

Applying the principle of "Power Tap", we should look for the most powerful way to make use of a divisibility condition. With a slight investigation, it is the most powerful in the following case: If b is a product of a few items, then having a being prime is the most powerful, as one of the items must be divisible by p . Same conclusion is not possible with composites. Therefore, in general it is useful to consider primes first in these functionals. If all prime values are confirmed, then we have infinitely known values. It is not difficult to confirm all non-prime values from here.

2. Secondly, we can interpret $a \mid b$ in terms of size. We have that $a \mid b$ implies $|a| \leq |b|$ or $b = 0$.

3 Easier Problems

1. (IMO HKTST 2020 Test 1 Problem 1) Find all function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every positive integer n the following is valid: If $d_1, d_2, \dots, d_s$ are all the positive divisors of n , then

$$f(d_1)f(d_2)\dots f(d_s) = n$$

2. (Balkan MO shortlist 2019 N1) Let $\mathbb{P}$ be the set of all prime numbers. Find all functions $f : \mathbb{P} \rightarrow \mathbb{P}$ such that

$$f(p)^{f(q)} + q^p = f(q)^{f(p)} + p^q$$

holds for all $p, q \in \mathbb{P}$.

3. (Balkan MO shortlist 2018 N2) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$n! + f(m)! \mid f(n)! + f(m!)$$

for all $m, n \in \mathbb{N}$

4. (Balkan MO shortlist 2017 N2) Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that the number $xf(x) + (f(y))^2 + 2xf(y)$ is a perfect square for all positive integers x, y .
5. (Balkan MO shortlist 2020 N3) Given an integer $k \geq 2$, determine all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(x_1)! + f(x_2)! + \dots + f(x_k)!$ is divisible by $x_1! + x_2! + \dots + x_k!$ for all positive integers $x_1, x_2, \dots, x_k$.

4 More Advanced Problems

- (5M) (IMO shortlist 2013 N1) Let $\mathbb{Z}_{>0}$ be the set of positive integers. Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that

$$m^2 + f(n) \mid mf(m) + n$$

for all positive integers m and n .

- (5M) (IMO shortlist 2012 N1) Call admissible a set A of integers that has the following property: If $x, y \in A$ (possibly $x = y$) then $x^2 + kxy + y^2 \in A$ for every integer k . Determine all pairs m, n of nonzero integers such that the only admissible set containing both m and n is the set of all integers.

(10M) (IMO shortlist 2019 N4) Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that $a + f(b)$ divides $a^2 + bf(a)$ for all positive integers a and b with $a + b > 2019$.

(15M) (IMO shortlist 2004 N3) Find all functions $f : \mathbb{N}^* \rightarrow \mathbb{N}^*$ satisfying

$$((f(m))^2 + f(n)) \mid (m^2 + n)^2$$

for any two positive integers m and n .

(20M) (IMO shortlist 2009 N3) Let f be a non-constant function from the set of positive integers into the set of positive integer, such that $a - b$ divides $f(a) - f(b)$ for all distinct positive integers a, b . Prove that there exist infinitely many primes p such that p divides $f(c)$ for some positive integer c .

(20M) (IMO shortlist 2016 N6) Denote by $\mathbb{N}$ the set of all positive integers. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all positive integers m and n , the integer $f(m) + f(n) - mn$ is nonzero and divides $mf(m) + nf(n)$.

(25M) (IMO shortlist 2007 N5) Find all surjective functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m+n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p .

(30M) (IMO shortlist 2011 N5) Let f be a function from the set of integers to the set of positive integers. Suppose that, for any two integers m and n , the difference $f(m) - f(n)$ is divisible by $f(m - n)$. Prove that, for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.

(30M) (IMO shortlist 2012 N5) For a nonnegative integer n define $\text{rad}(n) = 1$ if $n = 0$ or $n = 1$, and $\text{rad}(n) = p_1 p_2 \cdots p_k$ where $p_1 < p_2 < \cdots < p_k$ are all prime factors of n . Find all polynomials $f(x)$ with nonnegative integer coefficients such that $\text{rad}(f(n))$ divides $\text{rad}(f(n^{\text{rad}(n)}))$ for every nonnegative integer n .

(35M) (APMO 2021 P5) Determine all Functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(f(a) - b) + bf(2a)$ is a perfect square for all integers a and b .

(40M) (CMO 2020 P6) Find $f : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$, such that for any $x, y \in \mathbb{Z}_+$,

$$f(f(x) + y) \mid x + f(y).$$

(40M) (IMO shortlist 2013 N6) Determine all functions $f : \mathbb{Q} \rightarrow \mathbb{Z}$ satisfying

$$f\left(\frac{f(x) + a}{b}\right) = f\left(\frac{x + a}{b}\right)$$

for all $x \in \mathbb{Q}$, $a \in \mathbb{Z}$, and $b \in \mathbb{Z}_{>0}$. (Here, $\mathbb{Z}_{>0}$ denotes the set of positive integers.)

(40M) (IMO shortlist 2015 N6) Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Consider a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$. For any $m, n \in \mathbb{Z}_{>0}$ we write $f^n(m) = \underbrace{f(f(\dots f(m) \dots))}_n$. Suppose that f has the following two properties:

(i) if $m, n \in \mathbb{Z}_{>0}$, then $\frac{f^n(m) - m}{n} \in \mathbb{Z}_{>0}$; (ii) The set $\mathbb{Z}_{>0} \setminus \{f(n) \mid n \in \mathbb{Z}_{>0}\}$ is finite.

Prove that the sequence $f(1) - 1, f(2) - 2, f(3) - 3, \dots$ is periodic.

- (40M) (IMO shortlist 2015 N7) Let $\mathbb{Z}_{>0}$ denote the set of positive integers. For any positive integer k , a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ is called k -good if $\gcd(f(m) + n, f(n) + m) \leq k$ for all $m \neq n$. Find all k such that there exists a k -good function.
- (45M) (IMO shortlist 2018 N6) Let $f : \{1, 2, 3, \dots\} \rightarrow \{2, 3, \dots\}$ be a function such that $f(m+n) \mid f(m) + f(n)$ for all pairs m, n of positive integers. Prove that there exists a positive integer $c > 1$ which divides all values of f .